and find something else to do." He didn't have to because by January 2021 both Covishield and Covaxin were given emergency authorisation use by the Drugs Controller of India, a moment of great pride for both Ella and Poonawalla.

After that it was a question of ramping up their production lines and providing the numbers required for the vaccination drive run by the government while ensuring quality. The urgency for supplies soared after the second wave of Covid gutted the country with its spread and ferocity in the summer of 2021. The central government then moved to procure as many vaccines as possible with an ambitious target of inoculating every one of the 940 million adult population with double doses by the end of the year. They advanced the money to both manufacturers apart from encouraging other potential candidates to come out with vaccines. These included Zydus Cadila, which had taken the more advanced but untested route of using DNA strains to provide a vaccine that has recently been cleared for emergency use.



"BBIL and SII helped avert a health catastrophe and showed the world that India has the ability to make Covid vaccines that meet the highest standards"

DR V.K. PAUL
Niti Aayog Member

Meanwhile, two other Indian companies worked on novel ways to evolve a vaccine—Biological E Ltd tied up with US-based Baylor College of Medicine to develop a protein sub-unit vaccine called Corbevax; Gennova Biopharmaceuticals took the mRNA route that Moderna and Pfizer used to make vaccines. Corbevax has received clearance for emergency use while Gennova's vaccine is undergoing final trials. The central government had been in negotiations with multinational pharma giants like Moderna, Pfizer and Johnson & Johnson, but apart from higher costs and infrastructure requirements the deal breaker, Union health secretary Rajesh Bhushan says, was these companies insisting on a sovereign immunity waiver from India which would insulate them from lawsuits in case of failures or harm to those vaccinated. Other

the National Institute of Virology (NIV) helped us with quality control, standards and animal challenging studies. I have to salute all my employees, the government machinery, regulatory system and NIV. My employees worked day and night and that made the difference.

Q. You are now indigenously developing India's first nasal vaccine for Covid-19.
I call it the 'kiss' vaccine not just for fun. If somebody has taken this nasal vaccine and even kisses an infected person, s/he will not get Covid. Today, we are wearing masks to prevent transmission of the virus. My nasal vaccine will help stop transmission of the virus. Injectable vaccines cannot protect the upper respiratory tract, but the nasal drop will. Our internal studies show that it will also work as a booster dose for those who have taken Covishield or Covaxin.

Q. What is your advice for young entrepreneurs?
I want the younger generation to take risks and solve problems facing society. If you do that, you will be the best innovator, the best entrepreneur. Second is self-reliance, which Prime Minister Modi calls the Atmanirbhar Bharat progamme. Our success is an inspiration for others and proof that it can be done in the country against all odds. Third, going forward, they need to produce a generic vaccine– like the Pantavalent vaccine– that Unicef wants. Fourth is innovation. We did that with the rotavirus vaccine by making it an orally administered vaccine. Young entrepreneurs need to do this kind of breakthrough stuff. India should stand for innovation. Just copying drugs made by others will not help. It is innovation that will help generate the money to pave the way for further innovation.

Q. What do you think the Indian government should do to help?
While we are taking care of India's vaccine requirements, we should also be helping Africa, allotting 50 per cent of our vaccine production to them. There's this perception in Africa that Indian companies want to exploit African markets but they don't want to help solve Africa's problems. As Indian manufacturers, we have a moral responsibility to help Africa. I am working on a vaccine for cholera for Africa. Encouragement from the government, financial help, facilitation of clinical trials and regulatory process should give innovations fast-track clearances.

Q. Lastly, what drives you?
For me, everything–breathing, eating, talking–is directed towards science. I want to excel in it. There is nothing else for me but science.